

One-Switch Switchboard

TAPAN KUMAR MAHARANA

There is often a situation in industry when mounting too many switches is not possible and multiple loads need to be controlled with, as much as possible, very few switches. Presented here is a simple circuit that can control multiple loads with just one switch. Loads can even be turned on or off using this circuit.

Circuit and working

Fig. 1 shows the circuit for one-switch switchboard, which is built around

popular timer NE555 (IC1), decade counter CD4017 (IC2), quad AND gate CD4081 (IC3), two dual flip-flops CD4013 (IC4 and IC5), four transistors 2N2222 (T1 through T4), four 12V, 1C/O relays (RL1 through RL4) and some other easily-available components.

IC1 is wired as an astable multivibrator whose frequency can be varied using preset VR1. This frequency is fed to clock (pin 14) of decade counter IC2. The output of counter advances one count on receiving a pulse. Only four outputs (Q0 through Q3) have been used here and the fifth output (Q4) re-



sets the counter to start counting from Q0 again. The LEDs (LED1 through LED4) show status of the outputs of the counter.

The outputs of the counter are con-

Test Points

Test point	Details
TP0	0V, GND
TP1	12V when switch S1 is on
TP2	Frequency of astable multivibrator
TP3	High when switch S2 is pressed

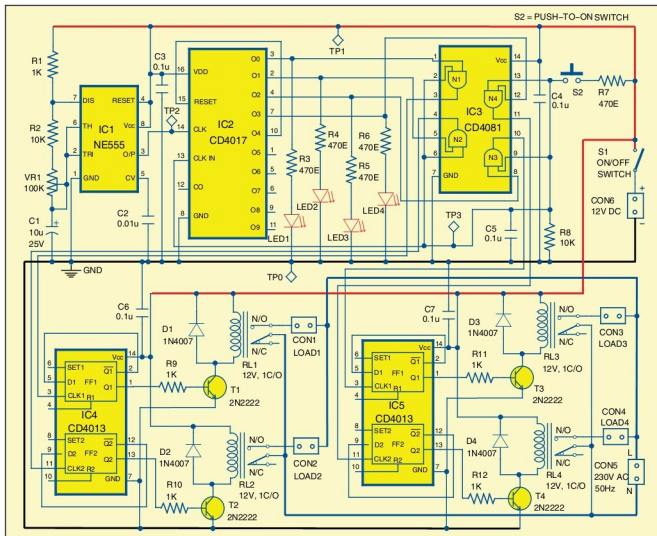


Fig. 1: Circuit diagram of one-switch switchboard